

MADRID



INTERNATIONAL CONFERENCE ON ACTIVE INFERENCE

ADVANCING THE FOUNDATIONS OF INTELLIGENCE

OCTOBER 14, 15 & 16, 2026

Spanish National Research Council (CSIC), Central Auditorium, Madrid, Spain

ACKNOWLEDGEMENTS

IWAI 2026 IS MADE POSSIBLE THANKS TO THE FOLLOWING PEOPLE

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WELCOME TO IWAI 2026

We are thrilled to welcome you to Madrid for the 7th edition of the IWAI International Conference on Active Inference. With this year's theme, Advancing the foundations of Intelligence, IWAI is a meeting point for interdisciplinary researchers who are trying to understand the underlying mechanisms of cognition and intelligence and developing new technology. After visiting Canada last year, the conference comes back to Europe, to Madrid, the capital of Spain.

With this framing, we have prepared a fantastic ensemble of speakers and debates ranging from the theoretical to the applied, in fields ranging from machine learning to robotics to computational neuroscience. This diversity reflects the cross-disciplinary impact of Active Inference and its potential to serve as a general theory of information processing.

We are also very excited to welcome you to Madrid. Madrid, first known as Mayrit (source of springs), is located at the centre of Spain, and it is a meeting point of different cultures, a very hospitable city that never stops, and a geographical and linguistic liaison between Europe, Africa and South America. Madrid was the residence of Santiago Ramón y Cajal when he was awarded the Nobel Prize in Physiology or Medicine. He started modern neuroscience under the neuron doctrine and influenced current breakthroughs in modern artificial intelligence. Thus, let us get some inspiration for the coming scientific future.

While we have a packed schedule laid out for you, we hope you get some time to explore the CSIC central and surroundings as well as the historical part of Madrid. This year we do not only show the power of Active Inference in research but also at the industry, displaying how this brain-inspired technology is slowly maturing.

Finally, the growing community of active inference has driven the IWAI transition from a workshop to an international conference. But we prepared IWAI as a conference that feels like a workshop. At IWAI, you can still meet, speak to, and connect with even the most senior thinkers in Active Inference. IWAI is an inclusive and respectful community. Take this opportunity - build connections, foster friendships and collaborations, and let's prepare for what comes next together!
Welcome to Madrid and to IWAI 2026!

Dr. Pablo Lanillos
Head of the Neuro AI and Robotics group, Spanish National Research Council
IWAI 2026 Chairs

PAST EDITIONS

VIRTUAL 2020 GHENT | 2021 BILBAO
ONSITE 2022 GRENOBLE | 2023 GHENT | 2024 OXFORD | 2025 MONTREAL

INVITED SPEAKERS

OPENING

Dr. José María Martell, Vice-President for Scientific and Technical Research (CSIC)

KEYNOTES

Prof. Rajesh Rao
Prof. Tadahiro Taniguchi
Prof. Giovanni Pezzulo
Prof. Carme Torras

FIRESIDE DEBATES

Action! The brain representation of behavior
Karl Friston, Rajesh Rao, Ruben Moreno Bote
Liset de la Prida.
Moderator: Ivilin Peev Stoianov

Ghost in the machine: AI-extended intelligence
Karl Friston, Sara Degli-Esposti.
Moderator: Pablo Lanillos

Are robotaxis ready?
Jorge Villagrá, Adam Bahlke, Johan Engström, Karl Friston.
Moderator: Martijn Wisse

LOCATIONS

Main Stage, Poster Room, Lunch

CSIC Central Auditorium
Calle de Serrano, 117
28006 Madrid

Fireside debates — CSIC, same building

Networking dinner

Comet Retiro
Calle de Alfonso XII, 30
28014 Madrid

IWAI 2026 LOCAL COMMITTEE
ABIAN TORRES TORRES. CAJAL NEUROSCIENCE CENTER, SPANISH NATIONAL RESEARCH COUNCIL, SPAIN.
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TOON VAN DE MAELE, THE BLANKET, UK

FOR MORE INFORMATION, VISIT THE EVENT WEBSITE

PROGRAM SCHEDULE

14
OCTOBER

WEDNESDAY 14 OCTOBER 2026 — NEURAL BASIS

08:30–08:55
REGISTRATION — AUDITORIUM CSIC

08:55–09:05
OPENING —
JOSÉ MARÍA MARTELL, VICE-PRESIDENT FOR SCIENTIFIC AND TECHNICAL RESEARCH (CSIC); PABLO LANILLOS AND IWAI ORGANIZERS

09:05–10:00
KEYNOTE: NEURAL BASIS (45'+10')
PROF. RAJESH RAO

10:00–10:30
COFFEE

10:30–11:30
SESSION 1 PREDICTIVE CODING AND REPRESENTATIONS
CHAIR: PABLO LANILLOS

11:30–12:00
SPOTLIGHT SESSION 1

12:00–13:00
FIRESIDE DEBATE: ACTION! THE BRAIN REPRESENTATION OF BEHAVIOR
RAJESH RAO (WASHINGTON), LISET DE LA PRIDA (CSIC), RUBEN MORENO BOTE (POMPEU FABRA), KARL FRISTON (UCL) MODERATOR: IVILIN PEEV STOIANOV

13:00–14:00
LUNCH

14:00–14:10
SPONSORS PRESENTATIONS

14:10–15:10
SESSION 2 MOTIVATION, INTEROCEPTION, AND AFFECT IN ACTIVE INFERENCE
CHAIR: MAHAULT ALBARRACIN

15:10–15:40
SPOTLIGHT SESSION 2

15:40–16:00
COFFEE

16:00–17:00
POSTER SESSION 1

17:00–19:30
— WELCOME DRINK AT CSIC

PROGRAM SCHEDULE

15 OCTOBER

THURSDAY 15 OCTOBER 2026 — AI AND COGNITIVE SCIENCE

08:30
— DOORS OPEN

09:05–10:00
KEYNOTE: ARTIFICIAL INTELLIGENCE (45'+10')
PROF. TADAHIRO TANIGUCHI. "COLLECTIVE PREDICTIVE CODING HYPOTHESIS: EMERGENCE OF LANGUAGE AND COOPERATIVE BEHAVIORS"

10:00–10:30
COFFEE

10:30–11:30
SESSION 3 MATHEMATICAL FOUNDATIONS OF ACTIVE INFERENCE
CHAIR: MARTIJN WISSE

11:30–12:00
SPOTLIGHT SESSION 3

12:00–13:00
FIRESIDE DEBATE: GHOST IN THE MACHINE: AI-EXTENDED INTELLIGENCE
KARL FRISTON (UCL) MODERATOR: PABLO LANILLOS

13:00–14:00
LUNCH

14:00–14:10
SPONSORS PRESENTATIONS

14:10–15:10
KEYNOTE: COG. MODELS (45'+10')
PROF. GIOVANNI PEZZULO. "FROM COGNITIVE MAPS TO SCHEMAS: ACTIVE INFERENCE FOR FLEXIBLE NAVIGATION"

15:10–15:40
SPOTLIGHT SESSION 4

15:40–16:00
COFFEE

16:00–17:00
SESSION 4 PLANNING, DECISION, AND CONTROL
CHAIR: IVILIN PEEV STOIANOV

17:00–18:00
POSTER SESSION 2

19:30–23:00
NETWORKING DINNER — COMET RETIRO
CALLE DE ALFONSO XII, 30, 28014 MADRID

PROGRAM SCHEDULE

16 OCTOBER

FRIDAY 16 OCTOBER 2026 — ROBOTICS AND APPLICATIONS

08:30
— DOORS OPEN

09:05–10:00
KEYNOTE: ROBOTICS (45'+10')
PROF. CARME TORRAS "HEALTHCARE ROBOTICS: RESEARCH CHALLENGES, CO-DESIGN, AND TECHNOETHICS EDUCATION"

10:00–10:30
COFFEE

10:30–11:30
SESSION 5 MULTI-AGENT, COLLECTIVE ACTIVE INFERENCE, AND ROBOTICS
CHAIR: HIDEAKI SHIMAZAKI

11:30–12:00
SPOTLIGHT SESSION 5

12:00–13:00
FIRESIDE DEBATE: ARE ROBOTAXIS READY?
JORGE VILLAGRÁ (CSIC), ADAM BAHLKE (MOTOR.AI), JOHAN ENGSTRÖM (WAYMO), KARL FRISTON (UCL) MODERATOR: MARTIJN WISSE.

13:00–14:00
LUNCH

14:00–14:10
SPONSORS PRESENTATIONS

14:10–15:10
SESSION 6 LANGUAGE, APPLICATIONS, AND PERSPECTIVES
CHAIR: DARIA DE TINGUY

15:10–15:40
SPOTLIGHT SESSION 6

15:40–16:00
COFFEE

16:00–17:00
POSTER SESSION 3

17:00–18:00
CLOSING REMARKS

CONTRIBUTED TALKS

October 14, 2026

1. PREDICTIVE CODING AND REPRESENTATIONS

10:30–11:30 • Chair: Pablo Lanillos

Error Highways: Scaling Predictive Coding to Very Deep Networks	Amirhossein Mohammadi, Alex Ororbia
Zero-Shot Representation Learning of Visual Sensory Stimuli through Laterally-Competitive Predictive Coding	Faezeh Habibi, Amirhossein Mohammadi, Alex Ororbia
Sensory Precision Inference for Multimodal Arbitration under Uncertainty	Tin Mišić, Takato Horii
Vision-Language Models as Gaussian Prior Preferences for Active Inference	Sensho Nobe

2. MOTIVATION, INTEROCEPTION, AND AFFECT IN ACTIVE INFERENCE

14:10–15:10 • Chair: Mahault Albarracin

Emotions as Deep Neuromorphic Priors in Self-Motivated Active Inference Agents	Elliot A. Giles, Paolo Pagliuca
Two routes to interoceptive prioritisation: dissociating likelihood precision and preference dynamics in active inference	St John Grimbly
Can Network Models of Science Do Without Pluralist Agents?	Jonas Hallgren, David Hyland, Mahault Albarracin
Belief-Based Maximum Occupancy Principle and Active Inference	Manolis Mylonas, Rubén Moreno-Bote

CONTRIBUTED TALKS

October 15, 2026

3. MATHEMATICAL FOUNDATIONS OF ACTIVE INFERENCE

10:30–11:30 • Chair: Martijn Wisse

Parameter Learning in Active Inference via Variational Message Passing	Benedikt Waldeck
Expected free energy as an information constraint on the Bethe Lagrangian	Wouter M. Kouw
Variational Laplace at Two Loops: Feynman Diagrams for the Variational Free Energy.	Stefano Goria
Renormalising Generative Models for Active Inference: Foundations, Derivations, and Verification	Karim Zaghw, Andrew Pashea, Marc Pritsch, Wouter W. L. Nuijten, Karl Friston, Lancelot Da Costa

4. PLANNING, DECISION, AND CONTROL

16:00–17:00 • Chair: Ivilin Peev Stoianov

Active Inference as a Convex Markov Decision Process	Nikola Milosevic, Nicolás Hinrichs, Nico Scherf
BeFree-Gate: Policy Composition under Partial Observability	Francesca Rossi, Giovanni Russo
Sophisticated Policies from Epistemic Priors	Wouter W. L. Nuijten, Bert de Vries
Active Inference with Quantum-like Cognitive Control	Catherine F. Higham

CONTRIBUTED TALKS

October 16, 2026

5. MULTI-AGENT, COLLECTIVE ACTIVE INFERENCE, AND ROBOTICS

10:30–11:30 • Chair: Hideaki Shimazaki

Collective sensing as emergent bayesian inference	Conor Heins
Expected Free Energy-based Informative Path Planning for Robotic Mars Exploration	Ajith Anil Meera, Pablo Lanillos, Wouter M. Kouw
From Environmental Scaffolding to Internalized Beliefs: Resilient Leader–Follower Configurations in Active-Inference Collectives	Alejandro Jimenez Rodriguez, Pranay Vaddepalli, Mahault Albarracin
Active Inference for Decentralised Traffic Coordination: A Multi-Agent Route-Choice Demonstration	Xue Yao, Dmitry Bagaev, Yanan Xin, Serge Hoogendoorn, Hans van Lint, Bert de Vries

6. LANGUAGE, APPLICATIONS, AND PERSPECTIVES

14:10–15:10 • Chair: Daria de Tinguy

One complex system organizing another: language signal structures hierarchical generative models across the lifespan	Evguenia Malaia
Balancing time and precision with active sensor selection and positioning	Esther G. van Pelt, Alexandru Onose, Ruud van Sloun, Harm Belt, Wouter M. Kouw
The clinician as affordance-architect: restoring breadth and depth of planning in persistent dizziness.	Marc A Broberg, Mick Thacker, Karl Friston, Jorge E Esteves
Transfinite Anatomy Theory: Selective Hierarchical Boundaries for Disease Modeling	Leon Moriguchi

POSTER SESSION 1

SPOTLIGHT SESSION 1

14 OCTOBER 2026

Primitive Representation by a Low-Dimensional Deterministic Bias on Predictive-Coding-Inspired Variational RNN	Takahito Fujimori, Jun Tani
Reverse engineering variational invariant structures in rodent brains	Takuya Isomura
Single State Update Predictive Coding training for Time Series Forecasting and Anomaly Detection	Matteo Cardoni, Sam Leroux
Exploring napping paradigm for Recurrent Spiking Neural Networks	Andreas Lie Massey, Stefano Nichele, Aliaksandr Hubin
A scale-invariant link between neural dynamics and active inference	Ukyo T. Tazawa, Ken Nakae, Teppei Ebina, Masanori Matsuzaki, Kenji Doya, Takuya Isomura
Precision-Adaptive Spiking Free Energy Networks for Control	Javier Zaragozano, André Rodrigues Urbano, Sander Wessel Keemink, Pablo Lanillos
Active, Actionable Scene Graphs	Pablo Bustos, Noé José Zapata Cornejo, Pedro Núñez
When Do Scene Beliefs Help Hard Attention? A Two-Stage Scene-Map Intervention Study	Pengcheng Pan, Yukie Nagai

POSTER SESSION 1

SPOTLIGHT SESSION 2

14 OCTOBER 2026

Neural Correlates of Mutual Information Beyond Uncertainty in Curiosity-Driven Exploration	Takuma Miyaguchi, Shin Nishimura, Kazutaka Ueda, Hideyoshi Yanagisawa
Arbitration Between Competing Needs and the Emergence of Goal Commitment in Active Inference	Matteo Priorelli, Giovanni Pezzulo
Partner-Specific Affective Precision in Social Active Inference	Harshil Shah, Andrew Pashea
Emotion in an active inference model of human driving	Julian F. Schumann, Johan Engstrom, Ran Wei, Jens Kober, Martijn Wisse, Arkady Zgonnikov
Conflict Resolution and Failure in Hierarchical Active Inference	Jaeseo Han
Principled Direction-Free Intrinsic Motivation through Model-Free Epistemic Free-Energy Estimators	Alireza Furutanpey, Schahram Dustdar
The Searchlight Test Revisited: An Active Inference Model Reproduces Disorganized Visual Exploration in Schizophrenia	Kazuki Yasuda, Hayato Idei, Hiroki Kojima, Ryosuke Izawa, Masashi Ikeda, Yuichi Yamashita
Cardiac Signals as Precision-Weighted Priors: An Active Inference Account of Heartbeat-Modulated Emotional Perception	Teresa Garzón Molina

POSTER SESSION 1

POSTER-ONLY

14 OCTOBER 2026

A Tutorial on Predictive Coding Networks	Björn van Zwol, Ro Jefferson, Egon L. van den Broek
Streaming Active Inference: A Problem Definition for Continuous Cognitive Runtimes	Zihang Wang; Boris Sedlak
Taxonomy of Mind Wandering and its Active Inference Formalization	Mengting Zhang
Putting the 'Body' in Embodied Cognition: An Active Inference Account of Concept Representation	Daniel Revach

POSTER SESSION 2

SPOTLIGHT SESSION 3

15 OCTOBER 2026

Expected Free Energy as Belief-Dependent Utility for p-POMDPs	Patrick Cooper
Constructing Markov Blankets from Actions.	Joséphine Pazem; Hans J Briegel, Lukas J. Fiderer
Physics of Information Geometry: Principle of Least Action on the Probability Simplex	Can Emre Koksal, Deniz Sargun
Uniqueness Maximization Active Inference for Encoder-Only Robotic Perception and Behavior Learning	Viet Dung Nguyen; Alex Ororbia
Distributional Active Inference	Abdullah Akgül, Gulcin Baykal, Manuel Haussmann, Mustafa Mert Çelikok, Melih Kandemir
Dröste Hierarchies in Predictive Processing: A Scale-Invariant, Non-Well-Founded Model of Recursive Selfhood in the Bayesian Brain	Burak Erdeniz, Aybars Yıldırım
Where Value Lives: A Comparative Survey of Reinforcement Learning, Control as Inference, and Active Inference	Emma Graham, Chelsea Joe, Daniel Ari Friedman
An On–Manifold Cost Repaid as Off–Manifold Robustness: Dissecting Information–Gain in Continuous–Control Active Inference	Stefano Goria

POSTER SESSION 2

SPOTLIGHT SESSION 4

15 OCTOBER 2026

Future shapes present: autonomous goal-directed and sensory-focused mode switching in a Bayesian allostatic network model	Hayato Idei, Jun Tani, Tetsuya Ogata, Yuichi Yamashita
Active Inference for Interaction-Mediated Control of a High-dimensional Robotic Arm	Fraser Paterson, Sebastian Stein, Markus Klar, John H. Williamson, Roderick Murray-Smith
Implicit Leadership Transitions in Shared Teleoperation via Free-Energy-Based Meta-Prior Adaptation	Hiroki Sawada, Shunichi Kasahara
Shared Goal Understanding in Physical Human-Robot Interaction through Deep Active Inference	Gabriel W. Haddon-Hill, Ferdinand Hartmann, Shingo Murata
Active Inference for Communication-Aware Update Selection in Interactive Holographic-Type Communication	Nika Hosseini, Martin Maier, Zakaria Abou el Houda
Benchmarking Generative Policy Proposals for Scalable Active-Inference Control	Yulin Li, Mohsen A Jafari, Andrea Matta
JAXbounce: A Continuous-Control Pong-Like Benchmark for Active Inference Beyond Low Dimensional Environments	Chaitanya Kaul, Markus Klar, Sebastian Stein, Fraser Paterson, John H. Williamson, Roderick Murray-Smith
Precision-Weighted Attentional Orienting: A POMDP Model of the Posner Cueing Task with Eye-Tracking Data	Luna Macías Salgado

POSTER SESSION 2

POSTER-ONLY

15 OCTOBER 2026

Amortizing Human Attention in Interactive Coding Agents: A Pilot Study	Pengcheng Pan; Xinfang Zhang
Understanding Integrative Complexity Through Active Inference	Dylan Grove; Alexander Hemming
The Rebundling Hypothesis: What LLMs Unbundle and Active Inference Could Restore	Stefano Goria
World Models: What They Are and What They Might Be	Rudolf Mester
Mental rehearsal as active inference over modularized generative models	Megumi Yoshihara; Takuya Isomura

POSTER SESSION 3

SPOTLIGHT SESSION 5

16 OCTOBER 2026

VAMAI: Variational Affordance Mapping and Active Inference for Training-free Autonomous Drone Navigation	Satyaki Maitra, Harshil Shah
Robotic Soccer Self-Localisation with active inference	Charel Van Hoof; Martijn Wisse, Noah van der Meer
Expected Free Energy Planning with Learned External-Camera Measurement Reliability	Joost Leliveld, Omur Arslan, Wouter M. Kouw
Deep Active Inference for Real-World Robotic Object Manipulation with Semantic Image Embeddings	Ko Igari, Kentaro Fujii, Shingo Murata
A Stochastic-Thermodynamic Active Inference Framework for Decentralized Swarm Control	Guido Quirijn Insinger
More Than Reward: Uncertainty Aware Ranked Sampling in Affordance-Conditioned Autonomous Tool Design	Abián Torres, Pablo Lanillos, Ricardo Sanz
Distributed Mine Search with Active Inference over Layout Hypotheses	Alexandre La Grappe, Daria de Tinguy, Emile LE FLECHER, Joao Borges de Sousa, Yoshiyuki Nishio
Hierarchical Active Inference and Meta-Learned World Models for Cooperative Multi-Agent Games	Alejandro Jimenez Rodriguez, Mahault Albarracin, Luca Manneschi

POSTER SESSION 3

SPOTLIGHT SESSION 6

16 OCTOBER 2026

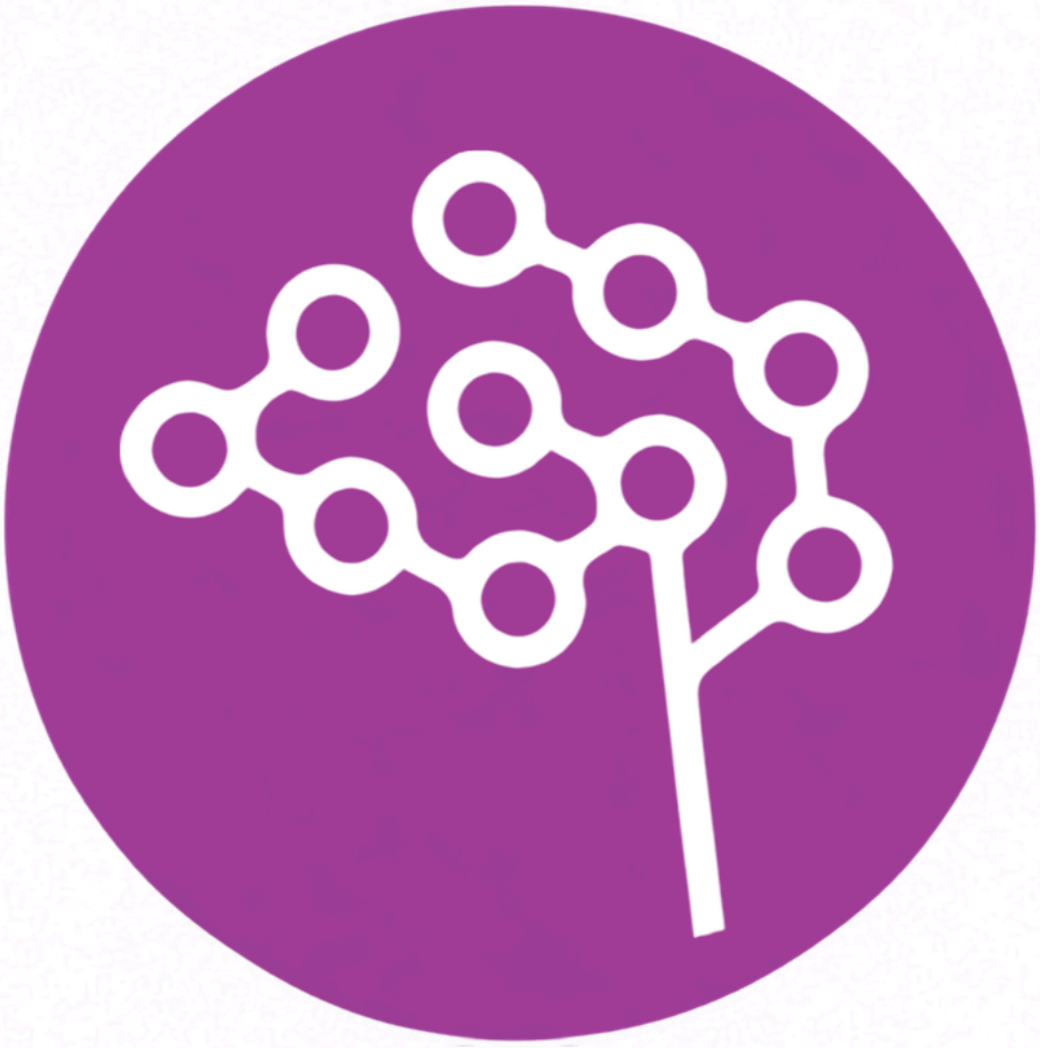
Asymmetric collaborative pricing via Active Inference and Mutual Modelling	Parth Patel, Kalliopi E Giannakopoulou, Ioanna PATSAROUCHA, Roderick Murray-Smith, Sebastian Stein
Speech that Protects a Model: A Type-Safe Active-Inference Engine for Compensatory-Defensive Discourse	Maria Svet
Hardware-Aware and High-Performance Message Passing for Efficient Active Inference Deployment	Nikola Pižurica, Matteo Risso, Nikola Milović, Alessio Burrello, Conor Heins, Miguel de Prado
Active Inference of Social Exclusion: Computational Modelling of Cyberball in Borderline Personality Disorder	Stefano Ventre; Rodolfo Rossi; Jialin Shi; Paolo Ossola
Active Inference as the Reasoning layer for Physical AI	Tim Verbelen, Toon Van de Maele, Corrado Pezzato, Riddhi J. Pitliya
Biological Bayesian Message Passing: A Brain-on-Chip with STDP-Based Training of hiPSC-Derived Neuronal Networks	Guillem Monsó, Wouter M. Kouw, Sepideh Adamiat
When Feedback Matters: An Active-Inference Model of the Speaking-Over-Writing Advantage in the Language-Production Effect	Roberto Limongi, Angelica Maria Silva
Framing Spatial Hearing within the Free Energy Principle: A Hierarchical Model of Perceptual Learning	Roberto Barumerli, Nils Meyer-Kahlen, Lorenzo Picinali

POSTER SESSION 3

POSTER-ONLY

16 OCTOBER 2026

Lexical Specificity through Message Passing in Multiscale Active Inference	Alexander Hemming; Dylan Grove
Belief Field Theory: Vector-Spin Collective Active Inference with Empirical Evidence	Thomas P. Kehler
Active Inference for Intention-Aware Human–Robot Collaboration Under Uncertainty	Shahriar Hassan; Pablo Bustos
Towards Adaptive Neuromorphic Sensing based on Active Efficient Coding	Quentin Halbach; Vibhu Kaushik, Aheli Saha, Nguyen H. P. Nguyen, Arne Roennau, Chiara Bartolozzi, Alessio Franci, Elisabetta Chicca, René Schuster; Claudia Lenk



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